ME46060 Engineering Optimization

Exercise 1 & 2

Course Code	ME46060	
Course Name	Engineering Optimization	
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Exercise 1

1.1 Analysis model

The Matlab script springanalysis1.m serves as a quantitative analysis model to evaluate a specific valve spring design. You feed it potential design variables (like mean diameter, wire diameter, and number of active coils) alongside fixed environmental parameters (like material properties, valve lift, and required pressures).

In return, the script calculates several performance and physical characteristics to see if the spring meets real-world constraints. Specifically, it calculates:

- **Physical Properties:** The volume, mass, and minimum compressed length (solid length plus clearance) of the spring.
- **Cost Estimates:** Material and manufacturing costs based on material weight and built-in volume.
- **Mechanical Performance:** The axial stiffness, the actual forces exerted at the closed and fully open valve positions, and the shear stresses experienced at those positions.
- **Dynamic Properties:** The first eigenfrequency of the spring to ensure it avoids resonance with the engine's cycle.

Based on a review of the MATLAB code and the provided references (Mott, Steinhilper, Roloff/Matek), here is how the theoretical relations have been implemented in the script:

- **Axial Spring Stiffness (k):** The script calculates stiffness as $k = \frac{G \cdot d^4}{8 \cdot D^3 \cdot n}$. This aligns directly with Steinhilper eq. (7.66) and Roloff/Matek eq. (10.48), and is an algebraic rearrangement of Mott eq. (7-6).
- **Blocking Length (L_b):** The blocking length (solid length) is implemented as $L_b = (n+nne) \cdot 1.02 \cdot d$. The 1.02 multiplier accounts for a 2% upper tolerance on the nominal wire diameter, which aligns with the principles for maximum solid length dimensions found in Steinhilper eq. (7.75/7.76) and Roloff/Matek eq. (10.40).
- **Clearance (S_a):** The required clearance between windings at maximum compression is calculated as $S_a = 1.5 \cdot \left(0.0015 \cdot \frac{D^2}{d} + 0.1 \cdot d\right) \cdot n$. This is an exact match for Roloff/Matek eq. (10.39), specifically the formula for dynamically loaded, cold-formed springs.
- **Spring Forces (F_1, F_2):** The forces at the closed and fully opened valve positions are calculated using Hooke's law, based on the deflection from the unloaded length (L_0). This corresponds to the general spring rate definitions found in Mott (p. 137) and Roloff/Matek eq. (10.49).
- **Wahl Factor and Shear Stress (τ_1, τ_2):** The script calculates a winding ratio $w = \frac{D}{d}$ and computes the Wahl factor using the polynomial approximation $1 + \frac{5}{4w} + \frac{7}{8w^2} + \frac{1}{w^3}$. This exactly matches Steinhilper eq. (7.61) [cite: 970]. The shear stresses are then calculated as $\tau = \text{Wahlfac} \cdot \frac{8 \cdot F \cdot D}{\pi \cdot d^3}$, which corresponds to Steinhilper eq. (7.60) [cite: 968], Mott eq. (7-4), and Roloff/Matek eq. (10.45a).
- **First Eigenfrequency** The lowest natural frequency is implemented as $freq1 = \frac{363 \cdot d}{n \cdot D^2}$. This maps to Roloff/Matek eq. (10.52), adjusted structurally for SI units (meters).

1.2 Optimization model

The design variables of the problem are presented in table 1.

Table 1: Characterization of Valve Spring Design Variables

Design Variable	**Symbol**	**Mathematical Type**	**Nature**
Mean winding diameter	D	Continuous	Deterministic
Wire diameter	d	Discrete	Stochastic
Unloaded spring length	L_0	Continuous	Deterministic
Number of active coils	n	Continuous	Deterministic

The design variables for the valve spring are characterized based on how they are selected and how they behave in a real-world manufacturing environment. Mathematically, Mean Diameter (D), Unloaded Length (L_0), and the Number of Coils (n) are treated as continuous because they can take any real value within their physical bounds to satisfy the stiffness (k) requirements. However, the Wire Diameter (d) is fundamentally discrete, as it is restricted to standard commercial gauges available from wire drawing mills.

Regarding their nature, most variables are modeled as deterministic, meaning we assume the value specified is exactly what is produced. However, in high-precision engineering like valvetrain design, these variables are actually stochastic. The wire diameter and material properties like G exhibit random variations due to manufacturing tolerances and batch inconsistencies. The analysis model acknowledges this by applying a 2% "buffer" to the wire diameter calculation for blocking length ($1.02 \cdot d$) and utilizing a safety factor for the natural frequency, ensuring the design remains robust even when slight random deviations occur.

Regarding the objective function, minimizing the manufacturing cost could be a possibility. This is a composite objective in the code that combines material mass (smass) and the space the spring occupies (bvol). This is proposed since most stakeholders value monetary benefits.
Alternatively, a Minimize Mass can also be an option by reducing valvetrain inertia, which is critical for high-performance engines to prevent "valve float."
These two objectives are likely conflicting since reducing the volume often requires using high-strength, expensive materials to handle the higher stresses of a compact design, or increasing the complexity of the winding, which raises the material cost.

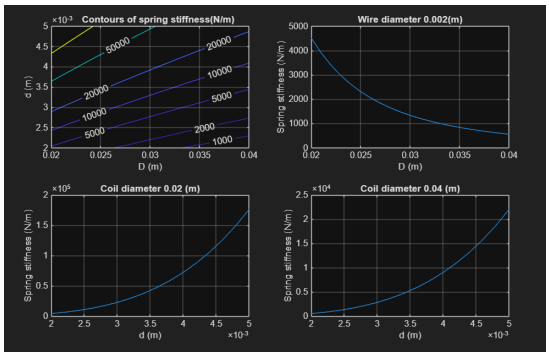
Finally, the constraints are shown below:

- In closed position, the spring force may not be lower than a given value, in order to prevent gas leakage.
- If the valve is in the open position, the spring force may not be lower than a given value, in order to prevent the rocker follower to come loose from the cam.
- Limits the maximum shear stress to prevent yielding or fatigue failure.
- The spring's natural frequency should be high enough to prevent resonance.
- L_2 should be larger than the blocking length.

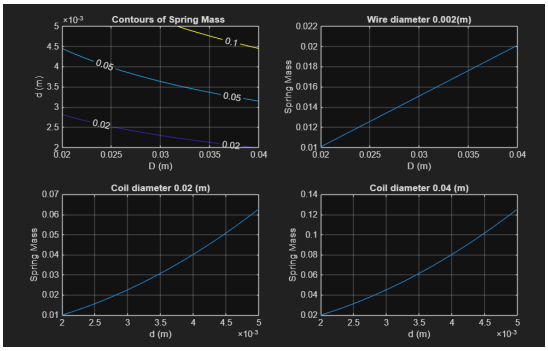
Exercise 2

2.1 Model Function Behavior

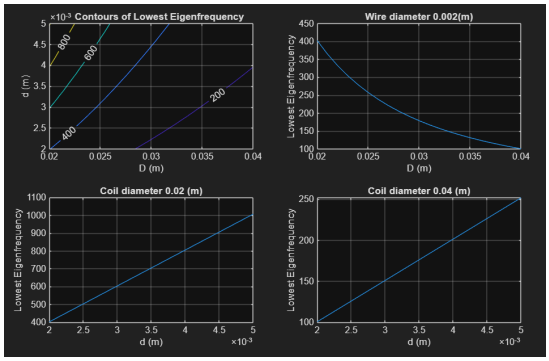
The axial spring stiffness k exhibits a highly non-linear behavior due to its mathematical formulation, where d is raised to the fourth power and D is cubed in the denominator. This relationship ensures the function is monotonically increasing with respect to the wire diameter d and monotonically decreasing with respect to the mean winding diameter D. Furthermore, the function is convex over the design space, a property evidenced by the contour lines "bowing" toward the origin and the non-linear, curved plots generated for constant values of d and D. The effects are demonstrated in Figure 1 and the key conclusions from the plots are summarized in Table 2



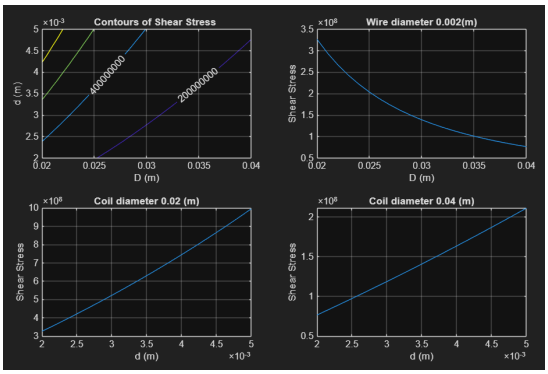
(a) Contours of Spring Stiffness



(b) Contours of Spring Mass



(c) Contours of Lowest Eigenfrequency



(d) Contours of Shear Stress

Figure 1: Behavioral analysis of valve spring design variables D and d.

Table 2: Summary of model function behavior for the valve spring design variables.

Variable	**Monotonicity** (D)	**Monotonicity** (d)	**Linearity**
k (Stiffness)	Decreasing	Increasing	Non-linear (d^4, D^{-3})
$smass$ (Mass)	Increasing	Increasing	Linear in D, Quadratic in d
$freq1$ (Frequency)	Decreasing	Increasing	Non-linear in D, Linear in d
τ_2 (Shear Stress)	Decreasing	Increasing	Highly Non-linear

2.2 Optimization Formulation

An optimization problem is posed. The objective is to minimize the mass of a spring by varying the mean winding diameter D and the wire diameter d. The constraints are written in negative null form with scaling as follows:

- $g_1 = \frac{L_{\min} - L_2}{L_{\min}} \leq 0$
- $g_2 = \frac{F_{1\,\min} - F_1}{F_{1\,\min}} \leq 0$
- $g_3 = \frac{F_{2\,\min} - F_2}{F_{2min}} \leq 0$
- $g_4 = \frac{Tau_2 - Tau_{12\,\max}}{Tau_{12\,\max}} \leq 0$
- $g_5 = \frac{freq1lb - freq1}{freq1lb} \leq 0$

and within the bounds $0.02 \leq D \leq 0.04$ and $0.002 \leq d \leq 0.005$.

Firstly, the limit value for the lowest eigenfrequency $freq1lb$ has to be determined. This is found by applying a safety factor $ncamfac = 13$ to the camshaft frequency f_{cam}. The latter is found using the following equation:

$$f_{\text{cam}} = \frac{n_{\text{cam}}}{60} = \frac{nm}{2 \cdot 60}, \tag{1}$$

where nm is the maximum engine speed in revolutions per minute, defined as 80 in the springparams1.m file. Hence $f_{\text{cam}} = 8.67$ Hz.

A file **springopt1.m** is set-up to compute the objective function and the constraint functions. A plot is then constructed to show the feasible region and the optimum point. This is shown in fig. 2.

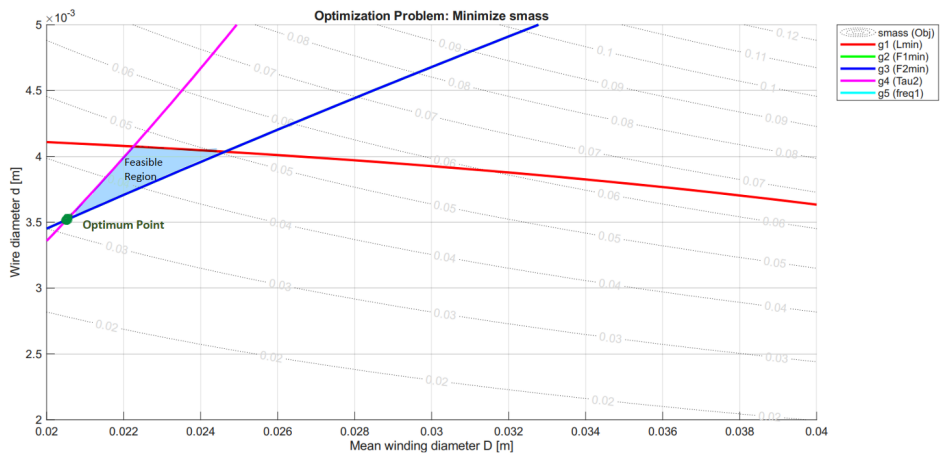


Figure 2: Plot of objective function contour with constraints

The feasible area is defined below the red line (as wire diameter d increases, the spring's solid length L_{min} increases, making it more likely to "block"), top left of the blue line (you need enough stiffness to maintain the minimum required force when the valve is open) and bottom right of the magenta line (increasing d reduces stress, but increasing F_2 (via d) increases it; physically, the stress limit prevents the wire from being too thin for the load.)

As can be seen, constraints g_2 and g_5 are not seen. This is because the computed value of stiffness is high enough within these boundaries for d and D, and the eigenfrequency only becomes an issue when D gets much larger in comparison to d, which is not the case in the current range. The active constraints are g_2 and g_3